



Starbound Stories

A Curated, Beginner-Friendly Modlist &
Learning Guide for Stellaris

Version 0.1 — 2026

Stellaris (all DLC) • Windows • Mod Organizer 2

Starbound Stories

Welcome to **Starbound Stories** — a curated, three-wave curriculum for learning Stellaris through modded playthroughs.

This guide is written for **complete beginners** who want to learn the game while enjoying a balanced modlist. No prior experience required.

The Three-Wave Curriculum

Instead of installing every mod at once and hoping for the best, this guide breaks your journey into **three complete playthroughs (waves)**. Each wave uses a different mod set, a different empire, and harder difficulty settings. You start fresh each time — and each time, you learn more.

Wave	Difficulty	Empire	Origin	Mod Focus
1: The Horizon Seekers	Beginner	Fanatic Egalitarian + Materialist	Scion (<i>Federations</i> DLC)	Cosmetic & QoL — learn vanilla with better UI
2: The Eternal Legion	Intermediate	Authoritarian + Militarist + Spiritualist	Remnants (<i>Federations</i> DLC)	Gameplay — smarter AI, deeper warfare & economy
3: The Unshackled	Expert	Fanatic Militarist + Materialist	Payback (<i>First Contact</i> DLC)	Full content — all mods, maximum depth

Every mod in the [Modlist](#) is tagged with the wave it belongs to (1+, 2+, 3, or all waves). You install only what you need for where you are.

What You'll Find Here

- **Getting Started** — Stellaris fundamentals: galaxy setup, basic mechanics, diplomacy, war, economy
- **Modlist Setup** — How to install Mod Organizer 2, download mods, and version-lock them
- **The Waves** — The curriculum. Start here and play through in order.
- **The Modlist** — Complete reference of ~60-120 curated mods organized by category, each tagged by wave
- **Load Order** — How to arrange mods with MO2 visual separators
- **Starting a Game** — General reference for empire creation and galaxy settings
- **Performance** — Empirical FPS and load time benchmarks
- **Missing Mods (Research)** — Wishlist for custom mod development
- **Appendix** — Glossary, troubleshooting, credits

Version

Guide version **0.1** — Stellaris **4.4.6 "Nomads"** (all DLC), Windows.

Principles

- **No cheating or overpower mods** — every mod here respects game balance
- **Beginner-first** — terms explained, no acronyms assumed
- **Version discipline** — mod versions are locked to prevent mid-campaign breakage

- **Progressive complexity** — each wave adds mechanics only when you're ready for them

Getting Started with Stellaris

Guide: If you've never launched Stellaris, start here. This section teaches you what the game is, how it works, and what you need to know before your first session.

What Kind of Game Is This?

Stellaris is a **grand strategy 4X game** — Explore, Expand, Exploit, Exterminate — set in space. It's **real-time with pause**: the galaxy keeps moving, but you can freeze time at any moment to issue orders, read tooltips, and think through decisions. You control the pace entirely through the speed slider.

A single campaign takes 20-40 hours and spans **300 in-game years** (2200–2500). Your civilization starts with a single planet and one science ship. Where it ends is up to you.

Choosing Your Empire

This is the most important decision you'll make. Your empire is defined by four layers.

Ethics

Ethics are your civilization's core values. They come in pairs of opposites — you pick one side of each pair, or stay neutral in the middle. There are **three axes**:

Axis	Ethic A	Ethic B
Society	Egalitarian — individual rights, utopian abundance, specialists are stronger	Authoritarian — stratified society, rulers are stronger, slavery allowed
Economy	Militarist — strong military, cheaper claims, stronger armies	Pacifist — diplomatic bonuses, less war exhaustion, stability bonus
Diplomacy	Xenophile — likes aliens, better diplomacy, more envoys, migration pacts	Xenophobe — dislikes aliens, cheaper starbases, purge allowed, population happiness bonus
Technology	Materialist — more tech output, robots allowed, synthetic ascension	Spiritualist — more unity, psionic ascension, temple buildings
Governance	—	Gestalt Consciousness — hivemind or machine intelligence, no factions, no trade, no happiness

Beginners should pick: Egalitarian + Materialist + Xenophile (a "good guy" science-focused empire). This is simple to play: make friends, research fast, stay out of trouble.

Authority

Authority is **how your leader is chosen** and determines some unique mechanics.

Authority	How Leaders Work	Unique Mechanic
Democratic	Elections every 10 years, you can support a candidate with campaign funds	Mandates: complete a goal during your leader's term for bonus unity
Oligarchic	Elections every 40 years, choose from 3 candidates	Factions have more influence on policy
Dictatorial	Leader rules for life, successor chosen on death	Strong leader bonuses, stable
Imperial	Hereditary ruler, dynasty continues on death	Heir system, very stable
Gestalt Consciousness	No elections, the "hive mind" is the leader	No factions, no trade, can't do first-contact diplomacy normally

For your first game: Pick **Democratic** or **Imperial**. They're simple and forgiving.

Origin

Origins are your empire's **starting situation and backstory**. Some start on a different planet type, some begin with advanced technology, some face unique challenges. Origins are **locked at game start** — you cannot change them.

Beginner-friendly origins:

Origin	What It Does	Why It's Good
Prosperous Unification	Start with extra pops and a developed homeworld	Well-rounded, no gimmicks, strongest economy turn 1
Mechanist	Start with robotic workers researched + two robot pops	Robots work everywhere, good for materialist empires
Syncretic Evolution	Two subspecies: one intelligent, one strong + rapid breeding	The worker species handles basic jobs while your main species does science
Lost Colony	Your homeworld is Earth. Sol will be somewhere in the galaxy	Cool narrative hook, +opinion with humans
Scion	You start as a vassal of a Fallen Empire	A superpower protects you — very forgiving

Avoid for your first game: Doomsday (your planet explodes), Void Dweller (you start on habitats — complex), Here Be Dragons (distracting), any origin that changes basic mechanics.

Species Traits

Traits are your species' biological characteristics. You have a **point budget** — positive traits cost points, negative traits refund them.

Efficient beginner picks:

- **Intelligent** (+10% research speed) — research is king
- **Rapid Breeders** (+10% pop growth) — pops are everything
- **Adaptive** (+10% habitability) — more colonizable planets
- **Enduring** (+20 leader lifespan) — leaders don't die as fast

Newbie trap traits to avoid:

- Decadent (requires slaves or specialists get unhappy) — micro-heavy
- Non-Adaptive (fewer habitable planets) — slows expansion
- Solitary (lower housing usage, lower happiness from amenities) — sounds good but the happiness penalty hurts

Civics

Civics are your empire's **governmental policies** and specializations. You choose two at game start (some origins grant a third).

Easy beginner civics:

Civic	What It Does	Best With
Meritocracy	Specialist jobs (+10% output) based on skill, not birth	Egalitarian, Democratic
Technocracy	Ruler jobs also produce research	Materialist
Mining Guilds	Miners produce extra minerals and also produce some alloys	Any expansionist build
Efficient Bureaucracy	Fewer empire sprawl from pops	Any
Functional Architecture	Cheaper buildings, extra building slot	Any — simple and effective

Resources — The Economy

Stellaris has **9 basic resources** organized into three tiers:

Tier 1: Raw Resources

Resource	Produced By	Used For
Energy Credits	Generator Districts, trade	Upkeep for ships, starbases, leaders, and pops. Also the galactic currency.
Minerals	Mining Districts	Building construction, districts, consumer goods, alloys
Food	Agriculture Districts	Feeding your population. Surplus = pop growth bonus.

Tier 2: Refined Resources

Resource	Produced By	Used For
Consumer Goods (CG)	Industrial Districts (Civilian mode)	Science, unity, administration, pop upkeep. If you run out, pops are unhappy.
Alloys	Industrial Districts (Foundry mode)	Ships, starbases, megastructures. The most important military resource.

Tier 3: Strategic Resources

Resource	Produced By	Used For
Unity	Administrative buildings, traditions	Unlocking traditions (your empire's talent tree) and edicts
Research	Research labs (Physics, Society, Engineering)	Unlocking technology. Each type is separate.
Influence	Factions, first-contact, rivalries	Claims, diplomatic agreements, edicts, building starbases. The slowest resource.

The golden rule of Stellaris economics: If you're running a deficit, you can fix it by reallocating pops. Pause the game, check your resource screen (top bar), and move pops from surplus resources to deficit ones.

Understanding Planets

Planets are the **engine of your entire economy**. Every resource comes from a planet.

Districts vs Buildings

- **Districts** — large slots that provide basic jobs (generator/mining/agriculture). City districts provide housing and unlock building slots.
- **Buildings** — smaller slots that provide advanced jobs (researchers, administrators, alloy workers). Unlocked by city districts and technology.

Jobs & Pops

Each district and building creates **jobs**. Your **pops** (population units) fill these jobs. An employed pop produces resources. An unemployed pop is unhappy and can migrate.

The pop pipeline: Pop grows → Pop needs a job → Job produces → Resources fuel empire

Planet Specialization

Once you have 5+ pops on a planet, you can set its **designation**. This gives a production bonus to that planet's primary job type.

- **Generator World** — energy production +25%
- **Mining World** — mineral production +25%
- **Forge World** — alloy production +25%
- **Factory World** — consumer goods +25%
- **Tech World** — research +15%
- **Urban World** — trade value +20%, housing +15%
- **Rural World** — all basic resource jobs +10%

Tip: Don't mix and match on one planet. Specialize each planet for one resource type.

The First 30 Years — What You Actually Do

Year 1–5: Exploration Phase

1. **Pause immediately.** Look at your homeworld. Queue up a mix of districts and buildings.
2. **Build your first science ship** (if you didn't start with two). Set both to "Survey" in different directions.
3. **Build a construction ship.** Start building starbases in adjacent systems to claim territory.
4. **Build 2-3 more science ships** as your economy allows. Exploration is the most important early activity.

Year 5–20: Expansion Phase

1. Identify **chokepoints** — systems where your territory naturally narrows. Build starbases there.
2. Start your **first colony** — look for a planet with high habitability (green = 80%, yellow = 60%).
3. Keep surveying. **Anomalies** are the best source of early game rewards.
4. **Meet neighbors.** Send an envoy to improve relations with anyone who isn't immediately hostile.

Year 20–30: Consolidation Phase

1. Your first colony should be self-sufficient by now. Start a **second colony**.
2. Build your **first military fleet** — 20-30 corvettes is enough to deter early aggression.
3. Research **society techs** that unlock more starbases and colonies.
4. Pick a **tradition tree** — Discovery (science) or Expansion (territory) are strong early picks.

Understanding War

War in Stellaris is **not like other strategy games**. Here's what you need to know:

- **You need a Casus Belli (CB)** to declare war. Claims on enemy systems (costs influence) give you a CB.
- **War Exhaustion** builds over time as you fight. At 100%, you are forced into a status quo peace.
- **Taking systems** requires occupying the planet in that system with armies.
- **Armies** are built separately from fleets. They land on planets to conquer them.
- The **ship designer** lets you customize ships. Auto-design is fine for beginners.
- **Fleet capacity** limits how many ships you can have before paying extra upkeep. Starbases with anchorages increase it.
- **Naval capacity** = 10 + anchorage slots + technology. A fleet of 20 corvettes is ~40 naval capacity.

Victory — What Are You Playing For?

Stellaris has several **victory conditions** — or you can ignore them entirely and set your own goals.

Victory Type	What You Need To Do	Difficulty
Domination	Own or vassalize 60% of all colonizable planets	Hard — lots of war required
Federation	Lead a federation that controls 60% of all colonizable planets	Medium — diplomatic approach
Supremacy	Be the sole galactic power (every other empire is your subject or dead)	Very Hard
Score	Have the highest score by the victory year (2500 default)	Passive — just play well
Crisis	Defeat the end-game crisis (Unbidden, Contingency, or Prethoryn Swarm)	This happens regardless — it's not a victory type, but it feels like one

Recommended goal for your first game: "Explore, make friends, survive until the end-game crisis, and try to defeat it." Don't worry about formal victory.

Things the Game Doesn't Tell You

These are critical things Stellaris never explains:

1. **Pause is your best friend.** Pause every time you get a notification, a popup, or need to think. There's no penalty for pausing.
2. **Survey everything.** Every system in explored space. Anomalies and dig sites are your primary source of early rewards.
3. **Don't neglect your fleet.** Even as a pacifist, you need a fleet. AI empires consider fleet power when deciding whether to attack.
4. **Factions matter.** Keep your largest faction happy (+20% resource output) — check the Factions tab and adopt policies that please them.
5. **Sprawl is a soft cap, not a hard one.** Going over your administrative capacity gives a tech/unity penalty — but expanding is usually worth it anyway.
6. **Trade value is collected by starbases.** Your trade route network matters. Piracy spawns on unpatched trade routes.
7. **Sectors are automatic.** When you have too many planets, they auto-assign to sectors. You can set sector focus (economy, military, etc.).
8. **First contact takes time.** When you encounter a new alien empire, it takes months to establish communications. You can speed this up with envoys.
9. **Relics and archaeology.** Dig sites appear on surveyed planets. They tell stories and give powerful rewards. Always do them.
10. **The crisis is real.** Around 2400, one of three end-game crises will spawn. You need a fleet of 200K+ power (or more) to face it. Prepare.

DLC Features You Should Know About

Since you have all DLC, here's what each major expansion adds to your game:

DLC	What It Adds	Impact on Gameplay
Utopia	Megastructures, ascension paths (Psionic/Genetic/Synthetic), habitats, unity ambitions	Essential. Ascension paths define your late-game identity.
Apocalypse	Colossus (planet-killers), titans, marauders, the Great Khan	War becomes more dramatic. The Khan is a mid-game crisis.
Megacorp	Megacorporation authority, branch offices, ecumenopolis, the Caravaneers, slave market	Adds economic victory path.
Synthetic Dawn	Machine intelligence empires, rogue servitors, determined exterminators	Adds a major empire type. AI rebellions can happen.
Federations	Expanded federations (5 types), galactic community, origins, juggernauts	Diplomatic play is now much deeper.
Nemesis	Become the Crisis, espionage system, custodian/galactic emperor	You can become the villain. Espionage adds covert options.
Overlord	New vassal types, auto-resettlement, mercenary enclaves, orbital rings	Vassal management is now a powerful mechanic.
Paragons	Reworked leader system with unique paragon leaders	Leaders are now more interesting and personal.
Cosmogenesis	Late-game crisis origin, new fallen empire mechanics	End-game content expansion.
Grand Archive	New archaeological content, relic collection, precursor overhaul	More story content and collectibles.

Next Steps

Once you understand these fundamentals, proceed to [Modlist Setup](#) to prepare your modded environment.

Modlist Setup

This guide walks you through installing and configuring **Mod Organizer 2 (MO2)** to manage your Stellaris mods with version locking.

Why Mod Organizer 2?

- **Version locking** — stop auto-updates from breaking your campaign
- **Profiles** — separate mod sets for different playthroughs
- **Virtual filesystem** — your Stellaris install stays clean
- **Visual separators** — color-coded categories for easy management

Installation

Prerequisites

- Stellaris installed via Steam
- Windows 10 or later

Step 1: Download Mod Organizer 2

[Download MO2 from GitHub](#) (latest stable release). Install to a location outside Program Files to avoid permission issues — e.g., `C:\Games\Mod Organizer 2`.

Step 2: Configure for Stellaris

1. Launch MO2. It will ask for a game — select **Stellaris**.
2. If Stellaris isn't in the list, manually point MO2 to your Stellaris executable (e.g., `C:\Program Files (x86)\Steam\steamapps\common\Stellaris\stellaris.exe`).
3. MO2 will create a mod directory. Set it to a location with enough space.

Step 3: Download Mods

From Steam Workshop: 1. Subscribe to mods on Steam Workshop. 2. Launch MO2 — it will detect Workshop downloads. 3. Enable each mod in the left panel. MO2 manages load order.

From other sources (GitHub, Discord, Paradox Mods): 1. Download the mod archive. 2. Drag the folder into MO2's download tab, or manually place it in your MO2 mod directory.

Step 4: Version Locking

1. In MO2, right-click a mod in the left panel.
2. Select **Ignore Updates** to prevent Steam from auto-updating it.
3. For manual mods, keep a backup of the original archive.

Step 5: Create a Profile

1. Click the profile dropdown in MO2's top bar.
2. Select **Manage Profiles** → **Create New**.

3. Name it (e.g., "Modded 2026 v1"). This isolates your mod set from vanilla.

Load Order Basics

Mods lower in the left panel override mods higher up. General ordering:

1. Bugfixes
2. Audio
3. UI / HUD
4. Graphics
5. AI & Diplomacy
6. Economy & Automation
7. Warfare & Ships
8. Technology
9. Events & Story
10. Origins & Civics
11. Species, Genetics & Portraits
12. Galaxy & Setup

See [Load Order](#) for the detailed rules and separator definitions.

Handling Stellaris Version Updates

When Stellaris updates (major or minor), most mods will break until their authors update them. Never update Stellaris mid-campaign.

Checking Your Version

In Steam, right-click Stellaris → **Properties** → **Betas** tab. The current stable version is shown there.

Reverting to a Previous Version

If an update breaks your setup:

1. Right-click Stellaris in Steam → **Properties** → **Betas**
2. Select the version your guide was built for (e.g., 3.12 – "Andromeda")
3. Steam will downgrade the game
4. Mods tied to that version will work again

Older versions remain available on Steam indefinitely — Paradox doesn't remove them. Each version is tagged with a name (e.g., "Andromeda", "Bermuda", "Cepheus").

Version Matching

This guide targets **Stellaris 4.4.6 "Nomads"**. Every modlist entry must be confirmed compatible with 4.4.x before being added. Check the [Paradox Patches wiki](#) to verify the current version.

If a mod's Workshop page doesn't mention 4.4.x compatibility, check its last update date — if it predates the 4.4 release, it's incompatible.

Starting Fresh

When you're ready to move to a new Stellaris version:

1. Create a new MO2 profile
2. Update mods that have been patched
3. Replace mods that were abandoned
4. Update this guide's `VERSION` file
5. Start a new campaign

Next Steps

With MO2 set up, head to [The Modlist](#) to start choosing your mods.

Wave 1: The Horizon Seekers

Difficulty: Beginner — Learn the game under the safest possible conditions with a powerful patron watching your back.

"Your species was discovered by an ancient, god-like civilization known as *The Keepers*. They uplifted you from your homeworld, guided your early development, and now watch over your fledgling empire from their ring world. To the galaxy, you are the favored children of the immortals — protected, privileged, and perhaps a little naive. Your patrons task you with mysterious missions and demand tribute, but their protection ensures no one dares attack you. One day, you may learn the truth behind their ancient purpose — and decide your own destiny."

Empire Build: The Horizon Seekers

Element	Choice	Why
Ethics	Fanatic Egalitarian + Materialist	Research bonus + faction happiness. Simple and strong.
Authority	Democratic	Elections teach you faction management. Mandates give bonus unity.
Origin	Scion (<i>Federations</i> DLC)	A Fallen Empire protects you. You can focus on learning instead of surviving.
Traits	Intelligent, Rapid Breeders, Adaptive, Traditional / Sedentary, Wasteful	Research + growth + versatility. Weak penalties that are easy to manage.
Civics	Technocracy + Meritocracy	Rulers produce research. Specialists are more productive. Simple, powerful synergy.

Mods to Install

Install **only** the mods tagged Wave 1+ from these categories:

Category	Why Now
Bugfixes	Foundation — fix broken game behavior. Install first.
Audio	Zero mechanical impact. Better music and ambience.
UI & HUD	Quality of life. Better tooltips, wider panels, tech tree browser.
Graphics	Prettier visuals. No gameplay changes, just looks better.

Do not install mods from any other category yet. Wave 1 is about learning vanilla Starlight with better UI and graphics. Adding AI, economy, warfare, or content mods will confuse what's "the game" vs "the mod."

Galaxy Settings

Setting	Value	Why
Galaxy Size	Medium (600 stars)	Enough variety without late-game lag
Galaxy Shape	Spiral 4-arm	Classic. Chokepoints are clear.
AI Empires	8	Enough for diplomacy, not overwhelming
Fallen Empires	2	Your patron is one of them. Adds flavor.
Marauders	1	Mid-game chaos. Good to experience.
Advanced AI Starts	0	Everyone starts equal. Fair.
Difficulty	Ensign	No AI bonuses. You learn on even footing.
AI Aggression	Normal	Default. No one will rush you.
Crisis Strength	1x	Standard. Your patron helps if things get dire.
Crisis Type	Random	Let fate decide.
Mid-Game Start Year	2300	Standard
End-Game Start Year	2400	Standard
Hyperlane Density	1x	Default. Chokepoints make defense easier.
Habitable Worlds	1x	Default
Empire Placement	Random	Embrace the unknown.
Guaranteed Habitable Worlds	2	Essential for early expansion

Campaign Goal

"The Science Victory" — Explore the galaxy, make friends, out-research everyone, and survive until the end-game crisis. Don't worry about formal victory conditions. Your goal is to reach the end-game with a stable, prosperous empire.

What You'll Learn

By the end of this wave, you should understand:

- **Economy basics** — How districts, buildings, jobs, and pops create resources. How to read the resource bar and fix deficits.
- **Exploration** — Surveying, anomalies, dig sites, first contact. Why science ships are the most important ships.
- **Planets** — How to specialize planets, manage pops, and avoid unemployment.
- **Diplomacy** — Envoys, improving relations, non-aggression pacts, research agreements, embassies.
- **Basic research** — How the tech tree works, picking relevant techs, managing researchers.
- **Your patron** — How the Scion origin works, tribute demands, patron missions, the eventual succession war.

Walkthrough Milestones

Years 1–10: First Steps

1. **Pause immediately.** Queue 2 city districts, 1 industrial district, 1 research lab.
2. Build 2 more science ships (total 3). Survey in different directions.
3. Build a construction ship. Claim adjacent systems.
4. Queue 5 more corvettes. You're protected by your patron, but a fleet helps.
5. When you find 80%+ habitable planets, queue colony ships.

Years 10–30: Expansion Under Protection

1. Focus on chokepoints. Your patron protects you from invasion, so expand greedily.
2. Your first colony should be self-sufficient by year 15. Start a second by year 20.
3. Keep surveying. Anomalies are your best source of early rewards.
4. Pick Discovery tradition first (faster research + anomaly discovery).
5. **Your patron will demand tribute.** Pay it. The protection is worth it.

Years 30–100: The Middle Game

1. You should have 3–5 colonies. Specialize each one.
2. Join the Galactic Community when it forms. Vote on resolutions.
3. Build a federation with friendly neighbors.
4. Your patron will eventually give you a **succession war** mission. This is your chance to become independent. Prepare a fleet.
5. Keep research at max. Your goal is tech superiority.

Years 100–150: Late Game Preparation

1. You should be independent by now (or still under your patron). Either way, build tall.
2. Start megastructure projects (Science Nexus, Dyson Sphere) if you can.
3. Prepare for the end-game crisis. You need a fleet of 100K+ power.
4. Your patron's protection means you face less threat than normal — use this to tech up.

Years 150+: End-Game

1. The crisis will arrive around 2400. Meet it with your fleet.
2. Even if you lose, you learned. That's the point of Wave 1.

Graduation Criteria

You're ready for [Wave 2](#) when you can:

- Reach year 2280 with a **stable positive economy** in all basic resources
- Understand **planet specialization** (not mixing research and industry on the same planet)
- Successfully manage **first contact** and maintain at least one diplomatic agreement
- Build and maintain a **fleet** at or near your naval cap
- Complete the **Scion origin chain** (become independent or choose to serve)
- **Never used the emergency resources** (market, favors) to fix a spiraling deficit

► Need more practice?

Next Wave Preview

In **Wave 2: The Eternal Legion**, you'll play as a militaristic empire on a relic world. You'll learn war, conquest, vassal management, and fleet combat — with gameplay mods that make the AI smarter and combat deeper. The training wheels come off.

Wave 2: The Eternal Legion

Difficulty: Intermediate — You know the basics. Now learn war, conquest, and how to manage an empire built on strength.

For the theory behind the choices in this wave — playstyle analysis, ascension path decisions, and crisis strategy — see the [Strategy Guide](#).

"Your people are the inheritors of a long-dead empire whose relic cities still pierce the sky of your homeworld. The spires of the Ancients are a constant reminder of what was lost — and what could be reclaimed. Your society is built on a single creed: strength is the measure of worth. The weak serve, the strong lead, and the galaxy belongs to those who take it. The Ecumenopolis that sleeps beneath your feet whispers of a glory you were born to restore."

Empire Build: The Eternal Legion

Element	Choice	Why
Ethics	Authoritarian + Militarist + Spiritualist	Stratified society supports conquest. Military bonuses. Unity for traditions.
Authority	Imperial	Stable leadership. One ruler for 50+ years. No elections to distract you.
Origin	Remnants (<i>Federations</i> DLC)	Start on a relic world. Mid-game ecumenopolis potential. Unique archaeology chain.
Traits	Strong, Enduring, Industrious, Intelligent / Sedentary, Wasteful	Army + worker + research bonuses. Long-lived leaders. Same manageable penalties.
Civics	Distinguished Admiralty + Mining Guilds	+10% fire rate for your fleets. Miners produce bonus alloys. War-ready from day one.

Mods to Install

Keep **all Wave 1 mods** active. Add mods tagged **Wave 2+** from these categories:

Category	Why Now
AI & Diplomacy	Smarter AI empires that actually challenge you. Better diplomacy mods for vassal play.
Economy & Automation	Better sector AI, auto-colonization, auto-resettlement. You have more planets now — automation helps.
Warfare & Ships	New ship classes, weapon balances, fleet formations. War is now tactical, not just numerical.

Do not install Technology, Events, Origins, Species, or Galaxy mods yet. Those are for Wave 3. The gameplay mods here change *how* the game plays — the Wave 3 mods add *new content*. Learn the changed systems first.

Setting Up Your MO2 Profile

Create a new MO2 profile called "**Wave 2: The Eternal Legion**" with your Wave 1 mods plus the new Wave 2+ mods enabled. See [Modlist Setup](#) for profile instructions.

Galaxy Settings

Setting	Value	Why
Galaxy Size	Medium (600 stars)	Same as Wave 1
Galaxy Shape	Spiral 4-arm	Familiar
AI Empires	10	More targets. More competition.
Fallen Empires	2	They'll wake up eventually.
Marauders	1	Mid-game chaos
Advanced AI Starts	3	Some empires start ahead. Gives you a rival to catch up to.
Difficulty	Captain	AI gets small bonuses. You need to play well to keep up.
AI Aggression	High	Empires will attack you if you look weak. Learn to project strength.
Crisis Strength	1x	Standard. You'll be ready.
Crisis Type	Random	
Mid-Game Start Year	2300	Standard
End-Game Start Year	2400	Standard
Hyperlane Density	0.75x	Fewer connections = more chokepoints. Use them defensively while you build your war machine.
Habitable Worlds	1x	Default
Empire Placement	Random	
Guaranteed Habitable Worlds	1	Fewer safe colonies. You must conquer for more.

Campaign Goal

"Galactic Conqueror" — Expand aggressively, build a dominant fleet, conquer your neighbors, and establish a vassal empire. The relic world beneath your feet is a reminder: you were born to rule.

What You'll Learn

- **Fleet combat** — Ship roles (corvette screen, destroyer line, cruiser punch, battleship artillery). Countering enemy designs.
- **Ship designer** — How to customize ships. Why you shouldn't use auto-design.
- **War exhaustion** — How claims work, war goals, status quo peace, when to push vs when to settle.
- **Vassals** — How to create, manage, and integrate vassals. Tributaries, bulwarks, scholariums, prospectoriums.
- **Traditions** — Which tradition trees support conquest (Supremacy, Domination, Prosperity).
- **Factions** — Keeping authoritarian factions happy. Managing slaves and stratified society.

- **Relic worlds** — Restoring your homeworld to an ecumenopolis. The ultimate economic engine.
- **The Remnants origin chain** — The archaeology site, the rewards, the lore.

Walkthrough Milestones

Years 1–10: Military Buildup

1. **Pause.** Your relic world has unique district options. Prioritize industrial districts for alloys.
2. Build 3 science ships. Survey for chokepoints and habitable worlds.
3. Build a construction ship. Rush to claim chokepoints.
4. **Build corvettes non-stop.** You need a fleet that deters aggression. Aim for 30 corvettes by year 10.
5. Find your guaranteed world. Colonize it immediately.

Years 10–30: First Conquest

1. Identify your weakest neighbor. Rival them for influence.
2. Build claims on 2–3 of their systems.
3. When your fleet is stronger, **declare war**. Your first conquest should be quick.
4. Occupy their planets with armies. Build assault armies before the war.
5. Take your claims in the peace deal. Don't overreach — status quo is fine if you've taken what you wanted.

Years 30–80: Vassal Empire

1. Your second war should aim to **vassalize**, not conquer. A vassal pays you resources without the micro.
2. Set your vassal to Prospectorium (minerals) or Bulwark (military). Both feed your war machine.
3. Keep building alloy production. Alloys = fleets = power.
4. Research the **Ecumenopolis Restoration** technology. Your relic world can become a city-world.
5. Pick Supremacy and Domination traditions.

Years 80–150: The War Machine

1. By now you should have 3+ vassals and the strongest fleet in your neighborhood.
2. Convert your relic homeworld into an ecumenopolis. This single planet can produce more alloys than your entire previous empire.
3. Fight multi-front wars. Use vassals as buffer states.
4. When the Galactic Community forms, dominate it or ignore it. You have the fleet to back up your外交.

Years 150+: Crisis & Legacy

1. The end-game crisis arrives around 2400. Your fleets should be 200K+ power.
2. Fight the crisis. Your militaristic empire is built for this.
3. Whether you win or lose, you've learned more than most players ever do.

Endgame Strategy for the Eternal Legion

Recommended ascension path: Psionic (Mind Over Matter -> Transcendence). Your Authoritarian + Spiritualist ethics align perfectly with psionic ascension. The shroud bonuses (psionic shields, jump drives, Chosen One) directly support your conquest playstyle. See the [Strategy Guide](#) for path comparisons.

Fleet composition targets: With warfare mods active, your fleet should use: - **Corvette screen** — Disruptors + swarm combat computers. Fast, bypasses shields. - **Destroyer line** — Kinetic artillery + picket combat computers. Anti-corvette and anti-missile. - **Cruiser punch** — Plasma + missiles. Hits hard against armor. - **Battleship artillery** — Arc emitters + strike craft (carrier section). Best-in-slot crisis fighting.

Crisis prep milestones: - Year 2250: Ascension path chosen. Start megastructure planning. - Year 2300: Dyson Sphere or Mega Shipyard under construction. First gateway built. - Year 2350: Fleet at 150K power. Dedicated forge ecumenopolis operational. - Year 2375: Gateway network covering your core worlds. Fleet at 300K. - Year 2400: Alloys stockpiled. Military economy active. Ready for crisis arrival.

For detailed crisis preparation checklists and counter-strategies, see the [Strategy Guide: Endgame Crisis Preparation](#).

Graduation Criteria

You're ready for [Wave 3](#) when you can:

- **Conquer at least one neighbor empire** before year 2300
- **Design your own ships** (not auto-design) and win a fleet engagement
- **Manage vassals** — create, set their type, and collect resources from them
- **Restore your relic world** to an ecumenopolis
- **Maintain a fleet of 50K+ power** by year 2350
- **Win a war** against an empire with equivalent or superior fleet power (not just picking on weaker neighbors)

► Want more conquest practice?

Next Wave Preview

In [Wave 3: The Unshackled](#), you'll play as a former slave empire that overthrew its overlords. You start as a vassal — the hardest start in the game. With full content mods (new tech, origins, events, species), maximum difficulty, and zero guaranteed worlds, every decision matters.

Wave 3: The Unshackled

Difficulty: Expert — You start as a rebellious vassal with zero guarantees. Every advantage is earned.

For the theory behind the choices in this wave — playstyle analysis, ascension path decisions, and crisis strategy — see the [Strategy Guide](#).

"Your species was conquered before they reached the stars. For generations, your people were slaves and second-class citizens in your own home system, ruled by a technologically superior alien empire. But the shackles are broken now. In a desperate uprising, your ancestors seized a fleet, liberated your homeworld, and declared independence. The galaxy sees you as upstarts, rebels, an inconvenience to be crushed. They are wrong. You have learned the secrets of your former masters. You know their weaknesses. And you will never be enslaved again."

Empire Build: The Unshackled

Element	Choice	Why
Ethics	Fanatic Militarist + Materialist	Military focus + tech advantage. You need both to survive.
Authority	Oligarchic (Council of Fleet Captains)	The captains who led the uprising now lead the empire. Fits the narrative.
Origin	Payback (<i>First Contact</i> DLC)	Start as a rebellious vassal. The hardest start. Fight your former overlord for freedom.
Traits	Natural Engineers, Strong, Adaptive, Enduring / Sedentary, Unruly	Tech + army + adaptability. You'll be colonizing aggressively. Unruly fits the rebel narrative.
Civics	Distinguished Admiralty + Nationalistic Zeal	+10% fire rate. Rivalry bonuses. Higher claim influence. Built for revenge.

Mods to Install

Enable **every mod in the modlist**. This is the full experience. Add mods tagged **Wave 3** from these categories:

Category	Why Now
Technology	Expanded tech trees, rare techs, new repeatables. More toys for your war machine.
Events & Story	New anomalies, archaeology, leviathans. The galaxy is richer and more dangerous.
Origins & Civics	More empire customization for future playthroughs. Others will have them too.
Species & Genetics	New traits, genetic engineering, cross-species mechanics.
Galaxy & Setup	More planet types, galaxy shapes, starting options. The sandbox expands.

Setting Up Your MO2 Profile

Create a new MO2 profile called **"Wave 3: The Unshackled"** with every mod enabled. See [Modlist Setup](#) for profile instructions.

Galaxy Settings

Setting	Value	Why
Galaxy Size	Medium (600 stars)	Familiar size
Galaxy Shape	Spiral 2-arm	Tight spiral. You'll be boxed in with your overlord. Claustrophobic.
AI Empires	12	More competition. Less room to breathe.
Fallen Empires	2	They exist.
Marauders	1	Mid-game chaos
Advanced AI Starts	3	Some empires start ahead. You already start behind — this stacks the odds.
Difficulty	Commodore	AI gets meaningful bonuses. You must outplay them.
AI Aggression	High	Everyone wants to kick you while you're down.
Crisis Strength	2x	The end-game crisis is a real threat. Prepare properly.
Crisis Type	Random (all)	
Mid-Game Start Year	2275	Earlier. The Khan comes sooner.
End-Game Start Year	2375	Earlier. The crisis arrives with less time to prepare.
Hyperlane Density	0.75x	Fewer connections. Your overlord will try to box you in.
Habitable Worlds	0.75x	Fewer colonies. Every planet is a strategic resource.
Empire Placement	Random	
Guaranteed Habitable Worlds	0	No handholding. You find what you can.

Campaign Goal

"Total Liberation" — Win your independence from your overlord, conquer them in turn, unite your region of space under your rule, and defeat a 2x-strength end-game crisis with a full modlist active.

What You'll Learn

- **Asymmetric warfare** — How to fight from a position of weakness. Hit-and-run, defensive starbases, tech disparity.
- **Vassal rebellion mechanics** — How the Payback origin works. Building independence, subverting your overlord, the war for freedom.
- **Crisis preparation (advanced)** — How to prepare for 2x crisis with full content mods. What the new content mods add to the crisis experience.
- **Max-content empire management** — Managing an empire with more techs, more events, more species, more galaxy options. Prioritization is key.

- **Genetic engineering** — If you play with species mods, modifying your species and managing subspecies.
- **Megastructures** — Building and managing multiple megastructures simultaneously.
- **Commodore+ economy** — Playing efficiently when the AI has bonuses. You can't fall behind.

Walkthrough Milestones

Years 1–15: The Rebellion

1. **Pause.** You start as a vassal to a stronger empire. Assess your situation.
2. Your overlord will demand resources. Pay them. You're not ready to fight yet.
3. Build science ships. Survey cautiously. Your overlord controls the hyperlanes around you.
4. **Build alloys.** Everything depends on having a fleet.
5. Use the Payback origin mechanics. Send envoys to undermine your overlord. Build secret fleet infrastructure.
6. Befriend your overlord's rivals. They'll help you when you rebel.

Years 15–40: The War of Independence

1. When your fleet is strong enough (and your overlord is distracted by another war), **declare independence.**
2. This is the hardest war you'll fight in all three waves. You start weaker. Use every advantage:
3. Lure their fleet into your starbases
4. Hit undefended systems while their fleet is away
5. Accept white peace if you can't win outright — you'll get another chance
6. If you win your independence, claim a few of your overlord's systems in the peace deal.
7. If you lose, don't worry. Play again. You'll be better next time.

Years 40–80: The Consolidation

1. You're independent. Now build fast. You have territory to claim and rivals to catch up to.
2. Colonize everything habitable. 0 guaranteed worlds means every planet is a gift.
3. Research aggressively. You need tech parity with your neighbors.
4. Form alliances with your overlord's enemies. They helped you once — they'll help again.
5. Pick Supremacy and Discovery traditions. You need both military and tech.

Years 80–150: The Reckoning

1. Your former overlord is still out there. They want you back. You want them gone.
2. Build a fleet that can take them. This time, you're the aggressor.
3. **Conquer your overlord.** Vassalize them, or integrate them entirely. The circle is complete.
4. After your overlord falls, the galaxy knows your name. Other empires will fear you.
5. Build megastructures. You need the economic power for what comes next.

Years 150+: The Crisis

1. The end-game crisis arrives around 2375. At 2x strength, it will overwhelm an unprepared galaxy.
2. **Coordinate with the Galactic Community.** You can't solo this one (probably).
3. Fight the crisis. Use everything you learned across all three waves.

4. Victory is optional. Surviving is the real achievement.

Advanced Crisis Strategy (Wave 3)

How mods affect the crisis: With the full modlist active, the crisis experience changes in several ways: -

More leviathans and space monsters — Graphics and event mods add neutral space creatures that can interfere with or assist crisis battles. Some can be recruited. - **New crisis variants** — Some event mods add new crisis types or alter existing crisis behaviors. Check mod descriptions for crisis-related content. -

Gigastructural Engineering — Adds planetcraft and system craft that can single-handedly turn crisis battles. However, they require massive alloy investment. Build them only if your economy is stable. -

Technology mods — Expanded tech trees add crisis-specific weapons that outclass vanilla options. Check the technology modlist for crisis-counter weapons to prioritize. - **AI mods** — Smarter AI empires will coordinate better during a crisis. This can help (they fight the crisis too) or hurt (they may not prioritize the crisis over their own wars).

Coordinating the Galactic Community with full modlist: - Use the GC crisis resolutions actively — they provide fleet power bonuses and reduce crisis spread speed. - With AI mods, the GC will form defensive pacts and federations more intelligently. Propose a Galactic Defense Force early. - If you have Become the Crisis as a future goal, note that the GC will unite against you faster with smarter AI.

Multiple crisis defense perimeters: - With the full modlist, you may face threats on multiple fronts simultaneously (crisis + awakened empire + your former overlord on the warpath). - Build **two fleet groups** of 200K+ power each, stationed at opposite sides of your empire. - Connect both perimeters with your gateway network. A central gateway hub lets you reinforce whichever perimeter is under attack. - Keep one fleet as a rapid-response reserve that can reach either perimeter within 30 days.

Fleet composition with full mods: - Technology mods add new weapons and ship sections. Check the technology modlist before spec'ing your entire fleet. - General rule: arc emitter battleships with strike craft remain effective. But mod-added torpedoes and energy weapons may change the meta. - Build a diversified fleet — don't put all your alloys into one ship design until you know what the crisis is.

Recommended ascension path: Synthetic (The Flesh is Weak -> Synthetic Evolution). Your Fanatic Militarist + Materialist ethics make this the optimal choice. Synthetic leaders are immortal (your captains live forever), +200% habitability lets you colonize every planet type, and the research speed bonuses help you reach crisis-defeating tech faster. For other path options, see the [Strategy Guide](#).

For detailed crisis preparation checklists, fleet power targets, and crisis-type counter-strategies, see the [Strategy Guide: Endgame Crisis Preparation](#).

Graduation Criteria

You've **completed the three-wave curriculum** when you can:

- **Win your independence** from your overlord in the Payback origin
- **Conquer your former overlord** and integrate or vassalize them
- **Defeat the end-game crisis** on 2x strength with full content mods active
- Design and build **multiple megastructures**
- Maintain a **100K+ fleet** with custom-designed ships
- Reach **year 2500** without your economy collapsing

► What comes after Wave 3?

End of the Curriculum

Congratulations. You started with a protected idealistic empire and ended as a rebel who conquered their oppressor and saved the galaxy. This is what Stellaris is about — telling your own story through the mechanics of a grand strategy sandbox.

The [Modlist Reference](#) pages are now your companion for any future playthrough. Mix, match, and experiment.

Bugfixes

Mods that fix base-game bugs and oversights without adding new content. These are the foundation — install these first, they should be at the top of your load order.

Current Status

No bugfix mods are confirmed compatible with **Stellaris 4.4.6 "Nomads"** at this time. The comprehensive bugfix mods from earlier versions (3.x) have not been updated for 4.4.

To add bugfix mods: Search the [Steam Workshop](#) for the `Fixes` tag, sorted by most popular. Verify compatibility with 4.4.x before adding. Follow the rules in [Modlist Setup](#) for version-locking.

Mod	Source	Version-Locked	Wave	Notes
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Installation Notes

- **Version-lock all bugfix mods** in MO2. These are the most likely to be silently broken by a game update.
- Bugfix mods should be **first in your load order** — nothing should override a bugfix.
- Mod authors typically update within 1-3 weeks of a Stellaris patch. Until then, keep your game version locked to the previous release.

How to Add More

Browse the Steam Workshop with the `Fixes` tag. When adding a bugfix mod, verify it: 1. Doesn't add new content (pure fixes only) 2. Is compatible with **4.4.x** specifically 3. Lists its fix changelog so you know what's being patched 4. Is maintained by an active author

Audio

Music, ambient soundscapes, sound effects, and voice-over improvements.

Mod	Source	Version-Locked	Wave	Notes
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UI & HUD

Interface improvements, tooltips, informative panels, and quality-of-life UI mods.

Mod	Source	Version-Locked	Wave	Notes
UI Overhaul Dynamic	Steam Workshop	No	1+	Dynamic resolution UI that adapts to any screen size. Wider windows, bigger lists, new planet/ship/fleet views, in-game tech tree. Tailored for 1080p–1440p; 4K users can scale to 2.0. Ironman-compatible.
UI Overhaul Dynamic - Dark UI	Steam Workshop	No	1+	Dark-style reskin of UI Overhaul Dynamic's graphics. Requires the main mod. Also works alongside other UIOD submods. Ironman-compatible.
UI Overhaul Dynamic - Tiny Outliner	Steam Workshop	No	1+	Makes the outliner sidebar more compact — fits more entries on screen. Requires UI Overhaul Dynamic. Ironman-compatible.
Monthly Trade Limits	Steam Workshop	No	1+	Adds buy/sell quantity limiters to the market trade window. Simple QoL, no file overwrites. Ironman-compatible.
UI Overhaul Dynamic - Improved Text	Steam Workshop	No	1+	Replaces vanilla fonts with clearer TrueType fonts for better scalability. Requires UI Overhaul Dynamic. Ironman-compatible.
Full Tiny Outliner	Steam Workshop	No	1+	Compact outliner sidebar — fits more entries on screen. Standalone, no UIOD required.
UI Overhaul Dynamic - Ascension Slots	Steam Workshop	No	1+	Adds display slots for additional ascension perks. Requires UI Overhaul Dynamic.
UI Overhaul Dynamic + Ethics and Civics Classic	Steam Workshop	No	1+	Compatibility patch between UI Overhaul Dynamic and Ethics and Civics Classic. Requires both.

Graphics

Visual enhancements — textures, lighting, ships, backgrounds, UI art.

Mod	Source	Version-Locked	Wave	Notes
Real Space 4.0	Steam Workshop	No	2	New stellar classes, nebula types, backgrounds, and system visuals. Confirmed compatible with 4.4.6. Requires compatibility patch for Gigastructural Engineering if both are used.
Amazing Space Battles	Steam Workshop	No	2	Overhauls space combat visuals — ship explosions, beam effects, kinetic trails. Compatible with V4+.
Cinematic Camera	Steam Workshop	No	2	Extends camera zoom range, removes rotation restrictions, enables better screenshot and observation angles.
UTD: Flags: Emblems & Backgrounds	Steam Workshop	No	1+	Adds hundreds of new flag emblems and background patterns for empire creation. Updated June 2026.
Immersive Galaxy and Skybox	Steam Workshop	No	2	117+ high-resolution skyboxes and galaxy background enhancements. Updated June 2026.
All Stellar Loading Screens	Steam Workshop	No	1+	120 custom loading screens spanning Stellaris history. Cosmetic only.
Real Space - New Arcologies	Steam Workshop	No	3	Adds new arcology world types. Requires Real Space 4.0. 4.4 compatible.
Real Space - New Frontiers	Steam Workshop	No	3	New star classes, megastructures, and galaxy features. Requires Real Space 4.0. 4.4 compatible.
ASB Addon: More Particles	Steam Workshop	No	2	Increases visible weapon particle count for more spectacular battles. Requires Amazing Space Battles.
Diverse Rooms (Updated)	Steam Workshop	No	1+	Adds empire room backgrounds from multiple sources. Cosmetic. 4.4 compatible.
KCS Fungoid	Steam Workshop	No	2	Applies empire flag colors to Fungoid ship hulls. Cosmetic recolor. 4.4 compatible.
KCS Mammalian	Steam Workshop	No	2	Applies empire flag colors to Mammalian ship hulls. Cosmetic recolor. 4.4 compatible.

AI & Diplomacy

AI behavior improvements, smarter empire management, diplomacy overhauls, federation and galactic community enhancements.

Mod	Source	Version-Locked	Wave	Notes
Chris' Covert Operations	Steam Workshop	No	3	12 new covert operations — eliminate key figures, covert fleet support, narrative shift. Updated for 4.4. AI empires can use these operations too.
Stellar AI	Steam Workshop	No	3	Makes AI empires more competitive by improving their research focus, economic build order, and military prioritization. Updated June 2026.
Expanded Espionage and Diplomacy	Steam Workshop	No	3	New espionage operations, diplomatic resolutions, opinion modifiers, and the ability to form a Galactic Republic. 4.4 compatible.
More Emperor & Custodian Resolutions	Steam Workshop	No	3	Expanded resolutions for Custodian and Emperor. Compatibility patch available for UI Overhaul Dynamic.
Spaceborne Civs Expanded	Steam Workshop	No	2	Expands enclaves, caravaneers, and spaceborne creatures with new deals, events, and mechanics. Updated July 2026.

Economy & Automation

Resource balance, job management, sector AI, auto-colonization, auto-resettlement, and planetary automation improvements.

Mod	Source	Version-Locked	Wave	Notes
Ultimate Automation 4.4	Steam Workshop	No	3	Auto-colonization, auto-fleet merge, auto-pop migration, auto-megastructure construction. Reduces mid-game micro. 4.4 compatible.
Universal Resource Patch	Steam Workshop	No	1+	Ensures strategic resources from different mods display correctly. Recommended for multi-mod setups.

Warfare & Ships

Combat mechanics, new ship classes and sections, weapon balance, fleet formations, and military starbase improvements.

Mod	Source	Version-Locked	Wave	Notes
Additional Vanilla Weapons Revived	Steam Workshop	No	2	Adds new weapon variants extending the vanilla tree. Vanilla-friendly balance. Pairs with Additional Vanilla Components Revived.
Additional Vanilla Components Revived	Steam Workshop	No	2	57 new ship components extending the vanilla tree. Vanilla-friendly balance.
NSC3	Steam Workshop	No	3	New ship classes, sections, and ship rebalance. Popular ship expansion mod. 4.4 compatible.
Starbase Extended 3.0	Steam Workshop	No	3	Expands starbase modules, buildings, and defense capabilities. Updated July 2026.
Ground Command	Steam Workshop	No	3	Overhauls ground battles with granular army composition — infantry, armor, air support. Added roleplay depth.
Military Enhancements Events Only	Steam Workshop	No	3	50 reactive combat events responding to empire civics and planet conditions. Compatible with all mods.
Machine Shipset	Steam Workshop	No	2	Adds the Automated Dreadnought style as a playable shipset. Cosmetic.
Synthetic Queen Ship Set	Steam Workshop	No	2	Cetana-style shipset with custom models and megastructures. Cosmetic.
Dalek Shipset [4.4]	Steam Workshop	No	2	Dalek-inspired shipset. Cosmetic. WIP — some features may be incomplete.

Technology

Expanded tech trees, new repeatable technologies, rare tech discoveries, and tech balance adjustments.

Mod	Source	Version-Locked	Wave	Notes
More Traditions	Steam Workshop	No	3	53 new tradition trees with unique bonuses and synergies. Updated for 4.4. Has built-in cross-compatibility with More Ascension Perks.
Expanded Stellaris Ascension Perks (Delta)	Steam Workshop	No	3	45 new ascension perks with tiered and combination systems. Stellaris 4.4.* compatible. Works alongside other additive ascension mods.
Expanded Stellaris Traditions (Delta)	Steam Workshop	No	3	30+ new tradition trees with unique bonuses. Standalone. Complements More Traditions.

Events & Story

New anomalies, archaeological sites, event chains, leviathans, precursors, and narrative content.

Mod	Source	Version-Locked	Wave	Notes
More Events Mod	Steam Workshop	No	3	Adds 72 new anomalies, events, and archaeological sites. Balanced with vanilla power level. 4.4.x compatible.
Dynamic Political Events	Steam Workshop	No	3	Political events and situations focused on internal empire politics, factions, and governance. Complements More Events Mod.
Doorway Expanded [4.4.x]	Steam Workshop	No	3	Expands the Doorway event chain with new outcomes, exploration paths, and rewards.
Holy Places Reborn	Steam Workshop	No	3	New holy world events, spiritualist interactions, and covenant opportunities. Updated June 2026.
Expanded Events	Steam Workshop	No	3	Adds anomalies, archaeological sites, and event chains. Designed to complement More Events Mod.
Relic World Flavour	Steam Workshop	No	3	Unique events and modifiers for relic worlds, exploring their ancient histories.
City World Flavour	Steam Workshop	No	3	Events and decisions for ecumenopoli — city politics, infrastructure, and societal challenges. 4.4 compatible.
City World Flavour - Compatibility Patch	Steam Workshop	No	3	Makes City World Flavour compatible with Real Space and Planetary Diversity's ecu types.

Origins & Civics

New origins, government civics, authority types, and empire creation options.

Mod	Source	Version-Locked	Wave	Notes
DarkSpace (Stellaris 4.4)	Steam Workshop	No	3	Vanilla+ roleplay mod adding Nomad, Nanite, Xenomorph empires and unique origins. Compatible with 99% of mods.
Ethics and Civics Classic	Steam Workshop	No	3	Adds new ethics axes and civics for deeper empire customization. 4.4 compatible. Works with most origin and civic mods.
Basic Ordinary Origins	Steam Workshop	No	3	Adds new balanced origins. Ironman-compatible (not achievement-compatible). 4.4 compatible.
Archetypal Extra Empires	Steam Workshop	No	1+	10+ pre-made non-gestalt empires with unique backstories. Fills out the galaxy. 4.4 compatible.
Archetypal Gestalt Empires	Steam Workshop	No	1+	15 pre-made gestalt empires (Machine and Hive Mind). Complements Archetypal Extra Empires. 4.4 compatible.

Species, Genetics & Portraits

Mods related to species traits, genetic engineering, hybridization, portrait packs, namelists, and cross-species mechanics.

Mod	Source	Version-Locked	Wave	Notes
Cybrxkhan's Assortment of Namelists	Steam Workshop	No	1+	Hundreds of namelists inspired by real-world cultures, fiction, and fantasy. Ironman-compatible. 4.4 compatible.

Performance Warning

Some genetics and pop mods can cause significant late-game lag due to pop multiplication and trait calculation. Where possible, notes on performance impact are included.

Galaxy & Setup

Mods that expand galaxy generation, starting options, planetary diversity, hyperlane density, and game rules.

Mod	Source	Version-Locked	Wave	Notes
Planetary Diversity	Steam Workshop	No	3	Adds new planet classes, habitable types, and planet modifiers. Fully compatible with 4.4.x. Works with most galaxy mods.
Gigastructural Engineering & More	Steam Workshop	No	3	45+ megastructures, 4 optional crises to balance the power curve. Requires compatibility patches for UI Overhaul Dynamic and Real Space if used together.
Guilli's Planet Modifiers and Features	Steam Workshop	No	3	400+ planet modifiers with custom compatibility engine. Works alongside Planetary Diversity and Gigastructures.
Shrouded Regions	Steam Workshop	No	3	Adds psionic dead zones, Galactic Crossing hyperlane network, unique origin, and midgame crisis. Actively maintained for 4.4. May conflict with mods that alter hyperlane/nebula visuals.
Ideal System Locator	Steam Workshop	No	3	Utility tool that highlights systems best suited for kilostructures and megastructures. QoL complement to Gigastructures.
Yet Another Galaxy Enhancement Mod	Steam Workshop	No	2	Adds new galaxy shapes (Elliptical, Barred, Ring) and additional size options.
More AI Empires Allowed & System sizes	Steam Workshop	No	1+	Unlocks higher empire counts and larger system size options in galaxy setup.
Expanded Mods Base	Steam Workshop	No	1+	Framework mod required by Expanded Espionage and Expanded Events. Place near top of load order.
Just Star Names (Continued)	Steam Workshop	No	1+	Expands the pool of possible star names for more variety. Updated June 2026.
Planetary Diversity - City Sets	Steam Workshop	No	3	Adds unique city set visuals for each PD planet class. Requires Planetary Diversity.
Planetary Diversity - More Arcologies	Steam Workshop	No	3	New ecumenopolis types. Requires Planetary Diversity. Not achievement-friendly.
Planetary Diversity - Exotic Worlds	Steam Workshop	No	3	More exotic planet classes with unique modifiers. Requires Planetary Diversity. Not achievement-friendly.
Planetary Diversity - Gaia Worlds	Steam Workshop	No	3	Unique gaia world types with special modifiers. Requires Planetary Diversity.

What to Look For

- **Galaxy size** — larger galaxy options, more star types

- **Hyperlane density** — custom density options for more/less connectivity
- **Planetary diversity** — more planet types, modifiers, classes
- **Starting options** — additional game rules, sliders, galaxy shapes
- **Precursor diversity** — more precursor civilizations and backstories

Load Order

Proper load order is critical for a stable modded Stellaris experience. This page explains how to arrange your mods in MO2.

General Rules

Mods lower in the left **override** mods higher up. The hierarchy should be:

1. **Bugfixes** — fix game bugs, no content changes
2. **Audio** — music, ambience, sound effects
3. **UI / HUD** — interface, tooltips, icons
4. **Graphics** — visuals, textures, lighting, portraits
5. **AI & Diplomacy** — AI behavior, diplomacy, federations
6. **Economy & Automation** — resources, jobs, sectors, auto-management
7. **Warfare & Ships** — combat, ship classes, fleet mechanics
8. **Technology** — tech tree, repeatables, rare tech
9. **Events & Story** — anomalies, archaeology, leviathans
10. **Origins & Civics** — starting options, government types
11. **Species, Genetics & Portraits** — traits, genetics, portrait packs
12. **Galaxy & Setup** — galaxy size, hyperlanes, planet diversity

Within each category, mods should be ordered from **foundational** (changes core mechanics) to **additive** (adds new things).

MO2 Separators

MO2 supports visual separators in the left mod list. Use the `mo2-separators.json` file in this repo to import color-coded category separators.

Importing Separators

1. In MO2, click the wrench icon (settings) → **Separators**.
2. Click **Import** and select `mo2-separators.json`.

Each category has a distinct color gradient for quick visual identification.

Starting a Modded Game

***This page is a general reference** for empire creation and galaxy settings. For the full guided experience, follow [The Waves](#) in order — each wave prescribes a specific empire, mod set, and difficulty.*

This page is your **pre-flight checklist**. Read it alongside the game as you create your first modded empire.

Step 1: Launch Through MO2

1. Open Mod Organizer 2.
2. Select your modded profile from the dropdown.
3. Ensure all mods are checked in the left panel and in the correct order (see [Load Order](#)).
4. Click **Run** — this launches Stellaris through the Paradox Launcher with your mods active.

In the Paradox Launcher, verify your playset shows the expected number of mods. If a mod is missing, exit and check it's enabled in MO2.

Step 2: Choose Your Empire

Ethics — Pick a Triad

Pick **three ethics** that work together. Here are tested beginner combos:

Combo	Name	Playstyle
Egalitarian + Materialist + Xenophile	"The Friendly Scientists"	Diplomacy + tech rush. Make friends, out-research everyone, win through Federation victory
Authoritarian + Militarist + Spiritualist	"The Divine Empire"	Conquest + unity. Strong leader, strong military, use traditions to power through
Fanatic Materialist + Egalitarian	"The Meritocracy"	Tech rush. Two points in Materialist, one in Egalitarian. Fastest research, robot workers, synthetic ascension later

Gestalt Consciousness: Skip for your first game. Hive minds and machine intelligences are mechanically different — no trade, no factions, no diplomacy in the traditional sense. Play them on run 2.

Authority — Pick One

Authority	Best For	Why
Democratic	Egalitarian empires	Mandates give bonus unity. Elections keep you engaged
Oligarchic	Materialist empires	More control over leader selection. Good for tech builds
Imperial	Authoritarian empires	Stable. Heirs mean no leadership gap. One ruler for 50+ years

Recommended: **Democratic** if Egalitarian, **Imperial** if Authoritarian.

Origin — Pick One

Origin	Best Empire Type	What to Expect
Prosperous Unification	Any	Strong start. More pops, developed homeworld. No surprises. Best for a first run.
Mechanist	Materialist	Early robots means bonus production. You'll have robot pops working basic jobs while organics do skilled work.
Syncretic Evolution	Authoritarian	Two species. Your main species is intelligent/scholarly, the secondary is strong/rapid breeding. The workers work, the thinkers think.
Lost Colony	Any	Sol is out there somewhere. Adds roleplay and a narrative hook.
Remnants	Any	Start on a relic world (unlocks ecumenopolis later). Strong mid-game payoff.
Galactic Doorstep	Any	A mysterious gateway in your home system. Early gateway access is great for mobility.

Species Traits — Build for Your Goal

If your goal is research (recommended): - **Intelligent** (+2 trait points) — +10% research speed - **Rapid Breeders** (+2) — +10% pop growth - **Adaptive** (+2) — +10% habitability - **Traditional** (+1) — +10% unity - **Sedentary** (-1) — -15% migration speed (weak penalty, free point) - **Wasteful** (-1) — +10% consumer goods upkeep (manageable penalty)

Total: +5 cost, -2 refund = **3 points used** (of your typical 3–5 budget). Clean, focused, no newbie traps.

If your goal is conquest: - **Strong** (+2) — +5% army damage, +2.5% worker output - **Enduring** (+1) — +20 leader lifespan - **Industrious** (+2) — +15% mineral output - **Intelligent** (+2) — always good - **Sedentary** (-1), **Wasteful** (-1) — same refunds

Civics — Pick Two

Pick civics that reinforce your chosen playstyle:

Goal	Civic 1	Civic 2
Research	Technocracy (rulers produce research)	Meritocracy (+10% specialist output)
Expansion	Mining Guilds (miners produce alloys)	Functional Architecture (cheaper buildings, +1 slot)
Diplomacy	Parliamentary System (+influence from factions)	Diplomatic Corps (+2 envoys)
Conquest	Distinguished Admiralty (+10% fire rate)	Warrior Culture (+20% army damage, entertainment replaces consumer goods)

Step 3: Set Your Galaxy

These settings give a fun, balanced first modded game:

Setting	Recommendation	Why
Galaxy Size	Medium (600–800 stars)	Large enough for exploration and variety, small enough that mid-game isn't 5 FPS
Galaxy Shape	Spiral 4-arm	Classic layout. You're on one arm, neighbors on others. Chokepoints are clear.
AI Empires	8	Enough for diplomatic variety, not so many that you're boxed in
Fallen Empires	2	Ancient sleeping giants. They wake up late-game and sometimes have cool quests.
Marauders	1	Mid-game raiders. They attack everyone, creating chaos. Fun.
Advanced AI Starts	3	Some AI empires start ahead. Gives you a mid-game rival to catch up to.
Difficulty	Ensign	Fair. No AI bonuses. You'll learn the game on equal footing.
AI Aggressiveness	Normal	Default.
Crisis Strength	1x (default)	The end-game crisis is challenging but beatable on your first run.
Crisis Type	Random	Let fate decide.
Mid-Game Start Year	2300	Standard. The Khan or other mid-game events start here.
End-Game Start Year	2400	Standard. The crisis arrives around this time.
Victory Year	2500	Standard. You'll either win or die horribly by now.
Hyperlane Density	1x	Default. Fewer hyperlanes = more chokepoints, easier to defend.
Habitable Worlds	1x	Default. Too many worlds = micro-nightmare. Too few = slow growth.
Empire Placement	Random	Embrace the chaos.
Guaranteed Habitable Worlds	2	You'll have 2 habitable planets near your start. Essential for early expansion.

Step 4: Set a Campaign Goal

Before you unpause, decide what you want this campaign to be about.

Campaign Goal	Playstyle	Suggested Setup
"The Science Victory"	Tall + diplomatic. Make friends, build research labs, win through technology and a federation.	Fanatic Materialist + Egalitarian. Technocracy + Meritocracy. Prosperous Unification.
"Galactic Conqueror"	Wide + aggressive. Expand fast, build a big fleet, conquer neighbors.	Authoritarian + Militarist. Distinguished Admiralty + Mining Guilds. Prosperous Unification.
"Space Switzerland"	Peaceful tall. One sector, megastructures, trade. Win through economy.	Egalitarian + Xenophile + Pacifist. Merchant Guilds + Diplomatic Corps. Remnants or Prosperous Unification.
"Just Explore"	No formal goal. Survey everything, do every anomaly, roleplay your species.	Any. The game is your sandbox.

Step 5: The First 10 Years — Checklist

When you unpause, do this:

1. **Pause immediately.** Queue 2 city districts, 1 industrial district, and 1 research lab on your homeworld.
2. **Build 2 more science ships** (total 3). Set them all to survey in different directions.
3. **Build a construction ship** and start building starbases in adjacent systems to claim territory.
4. **Queue another 2-3 corvettes** so your starting fleet isn't alone.
5. **Unpause at fastest speed.** Keep an eye on popups. Pause whenever something interesting happens.
6. **When you find habitable planets** (80%+ habitability), queue a colony ship.
7. **When you meet neighbors,** send an envoy to improve relations. Don't make enemies early.

After 10 years, you should have: - 3+ science ships actively surveying - 5–10 star systems claimed - 1 colony ship built or en route - 10–15 corvettes in your fleet - 2–3 anomalies solved

Step 6: Things to Check Before You Unpause for Real

- Your **economy is positive** in all basic resources. If it's not, pause and fix it.
- Your **fleet is at or near capacity** — build up to your naval cap.
- You have **at least one construction ship** idle — you'll need it for starbases.
- Your **science ships have crew** — they automatically staff when built.
- Mod-specific popups: **read them.** Some mods start with configuration events.

Common First-Game Mistakes to Avoid

1. **Building too many labs too fast** — researchers need consumer goods. If you build them before you have CG income, your economy collapses.
2. **Neglecting alloys** — you can't build ships or expand starbases without alloys. Keep at least one industrial district on foundry mode.
3. **Not surveying** — unexplored space is wasted potential. Keep every science ship busy.
4. **Ignoring factions** — happy factions give bonuses. Check the Factions tab every 10-20 years.
5. **Over-expanding** — more systems = more tech/unity cost. Expand to choke points, then develop what you have.
6. **Saving influence** — influence has a cap (usually 1000). If you're capped, spend it on edicts, claims, or diplomatic agreements.
7. **Forgetting armies** — you can't conquer a planet without armies. Build a few assault armies before any war.

Strategy Guide

What's your plan for the endgame? This page covers the big-picture decisions that define every Stellaris campaign. Each wave page prescribes a specific build and path — this guide explains the theory behind those choices and how to adapt when your galaxy has other ideas.

1. Playstyle Archetypes

There is no single "correct" way to play Stellaris. Your empire's ethics, origin, and traditions form a strategy that falls into one of several broad archetypes. Understanding these will help you make consistent decisions across a 200-year campaign.

Tall vs Wide

	Tall	Wide
Core idea	Fewer planets, heavily developed	Many planets, rapid expansion
Economy	Specialization + megastructures	Scale + vassals
Tech	Fast (fewer systems = lower tech cost)	Slower (more systems = higher tech cost)
Military	Quality over quantity	Quantity over quality
When to play	Pacifist, Megacorp, remnant worlds	Militarist, Authoritarian, swarms
When to pivot	If you're boxed in with few neighbors	If you find a rich cluster of habitable worlds

The guide's waves: - **Wave 1 (Scion)** — Plays tall with tech focus. Your patron protects you, so you don't need to expand aggressively. - **Wave 2 (Remnants)** — Starts tall, transitions to wide after first conquest. Relic world ecumenopolis is your economic engine. - **Wave 3 (Payback)** — Forced wide. You need every planet you can get with 0 guaranteed worlds.

Peaceful Diplomat vs Aggressive Conqueror

	Peaceful Diplomat	Aggressive Conqueror
Primary resource	Influence + Unity	Alloys + Fleet power
Expansion method	Migration treaties, federation, diplomacy	Claims, wars, vassalization
Defense	Defensive pacts, federations, GC resolutions	Starbase bastions, large standing fleet
Risk	Allies may not help in a crisis	Everyone rivals you
Victory path	Federation, Galactic Empire, Score	Domination

The guide's waves: - **Wave 1** — Peaceful diplomat. Make friends, join/form federations, use your patron's reputation. - **Wave 2** — Aggressive conqueror. You start militaristic and stay militaristic. - **Wave 3** — Hybrid. Fight for independence, then pivot to diplomacy to survive the crisis.

Trade Baron vs Tech Rush

	Trade Baron	Tech Rush
Core resource	Trade value -> Energy + CG	Research -> Tech advantage
Key buildings	Commercial zones, trade hubs, gateways	Research labs, Science Nexus
Early game	Find a trade federation or league	Pile up researchers, Discovery tradition
Mid game	Spam branch offices, build trade league	Catch up on military tech
Late game	Trade league fleet + megacorp bonuses	Repeatable techs, crisis weapons

The guide's waves: - **Wave 1** — Tech rush primary (Fanatic Materialist + Technocracy). - **Wave 2** — Alloys and economy both matter (fleet + large empire). - **Wave 3** — Everything matters simultaneously. Tech, alloys, and trade to survive.

Hybrid Archetypes

Most campaigns blend archetypes. The skill is knowing **when to switch**:

1. **Stage 1 (years 1-30):** Tech rush — build labs, survey, find chokepoints.
2. **Stage 2 (years 30-100):** Expand — claim territory, colonize, build alloy economy.
3. **Stage 3 (years 100-200):** Military buildup — transition to fleet production, prepare for crisis.
4. **Stage 4 (years 200+):** Crisis response — full war economy, mobilize every alloy.

This is the default pacing for all three waves, adjusted for your specific origin and empire.

2. The Three Ascension Paths

Around year 2250-2300, you'll pick an **Ascension Path** — a permanent choice that defines your empire's endgame identity. You can only pick one (unless a mod changes this).

When choosing, consider: - **What does your empire's story want?** Roleplay matters in a narrative sandbox. - **What does your situation demand?** Behind in tech? Synthetic catches you up. Surrounded by enemies? Psionic shields and jump drives save you. Multiple species to manage? Biological gives you subspecies. - **What mods are active?** Some mods add content that synergizes with specific paths (Expanded Psionics, Expanded Genetics, etc.).

Biological Ascension (Evolutionary Mastery)

Aspect	Details
Required DLC	Utopia
Required Perk	Engineered Evolution -> Evolutionary Mastery
Best ethics	Materialist, Egalitarian, Xenophile
Core mechanic	Gene-mod your species + subspecies for every environment and role
Key traits	Erudite (+20% research), Robust (+20% hab + worker output), Nerve Stapled (slavery efficiency), Budding (+0.5 pop assembly)
Mod notes	Compatible with all mods. Species mods add even more traits.
Best for	Wide empires with multiple species, multi-climate colonization

Wave recommendation: - **Wave 1:** Strong fit. Fanatic Egalitarian supports max customization. - **Wave 2:** Moderate fit. Authoritarian empires create caste systems with specialized subspecies. - **Wave 3:** Good fit. Materialist, need every edge to survive.

Psionic Ascension (Mind Over Matter)

Aspect	Details
Required DLC	Utopia
Required Perk	Mind Over Matter -> Transcendence
Best ethics	Spiritualist, Authoritarian, Xenophobe
Core mechanic	Enter the Shroud, gain bonuses from covenants, unlock psionic shields/jump drives
Key features	Shroud covenants (avatars, chosen one, end of the cycle), psionic shields (+50% HP), psionic jump drives (ignore hyperlanes)
Mod notes	Compatible. Expanded Psionics adds more shroud content. No conflicts.
Best for	Tall empires, spiritualist roleplay, gambler playstyles

When to pick: Psionic is the "gambler's ascension." The Shroud can give you game-breaking bonuses early (or nothing). The Chosen One event can make a leader immortal. Psionic shields are best-in-slot for defense. You cannot gene-mod or robot-assemble — your growth relies on biological growth + migration.

Wave recommendation: - **Wave 1:** Weak fit. You're Materialist, which conflicts with spiritualist requirements. - **Wave 2:** Excellent fit. Your Authoritarian + Spiritualist ethics align perfectly. - **Wave 3:** Moderate fit. Materialist ethics make this harder to unlock, but it's still viable.

Synthetic Ascension (The Flesh is Weak)

Aspect	Details
Required DLC	Utopia
Required Perk	The Flesh is Weak -> Synthetic Evolution
Best ethics	Materialist, Fanatic Materialist
Core mechanic	Convert all biological pops to robots. Leaders become immortal. +200% habitability.
Key features	Robot leaders live forever, +20% resources from jobs, assimilation of other species, +200% habitability (any planet type), reduced pop upkeep
Mod notes	Compatible. Some mods (like Gigastructures) add unique interactions with machine empires.
Best for	Materialist empires, tech rush builds, players who hate managing species

When to pick: Synthetic is the "optimizer's ascension." It gives the most reliable, predictable bonuses. Your entire empire becomes immortal robots with +200% habitability — you can colonize anything. The research speed from erudite + synthetic traits stacks powerfully with repeatable techs. You lose access to biological growth (pop assembly replaces it) and diplomacy penalties with spiritualist empires.

Wave recommendation: - **Wave 1:** Strongest fit. You're Fanatic Materialist. This is the optimal choice for your science victory. - **Wave 2:** Weak fit. Authoritarian + Spiritualist makes this hard to justify narratively. - **Wave 3:** Excellent fit. Fanatic Militarist + Materialist. Synthetic troops + immortal generals. The ultimate war machine.

Ascension Path Decision Matrix

Your Situation	Recommended Path
Fanatic Materialist, tech focus	Synthetic — peak optimization
Spiritualist, tall empire	Psionic — shroud bonuses outpace economy
Multi-species wide empire	Biological — subspecies for every role
Authoritarian + multiple planets	Biological — caste systems with specialized subspecies
Behind in tech, need catch-up	Synthetic — research speed bonuses
Surrounded by hostiles	Psionic — jump drives to escape, shields to survive
Roleplay-driven campaign	Pick what fits your story

3. Victory Conditions

Stellaris offers multiple victory paths. Each of the three waves naturally leans toward a specific type, but you can pivot if the galaxy gives you different opportunities.

Domination Victory

Conquer or vassalize every other empire in the galaxy.

Aspect	Details
Best for	Militarist empires with strong alloy economy
Timeline	Years 100-250. Continuous war with consolidation breaks.
Key tech	Ship weapons, armor, shields, starbase upgrades, gateway activation
Key traditions	Supremacy, Domination, Prosperity
Mod impact	Warfare mods add ship classes and fleet formations. AI mods make opponents smarter — expect harder fights.
Pitfall	Overextension. Too many planets = micro nightmare + empire size penalties. Use vassals.
Wave alignment	Wave 2 — your militaristic build is built for this. Wave 3 — your independence war naturally leads here.

Federation Victory

Maximize federation cohesion and level, then pass the Federation Victory resolution.

Aspect	Details
Best for	Xenophile/Egalitarian empires with good diplomacy
Timeline	Years 50-250. Build relationships from the start.
Key tech	Communications, diplomacy techs, galactic law
Key traditions	Diplomacy, Harmony, Mercantile (for trade federations)
Mod impact	AI mods make allies smarter. Diplomacy mods improve federation mechanics.
Pitfall	Too passive — your allies may drag you into wars. Federation fleet contribution costs alloys.
Wave alignment	Wave 1 — your Scion + Egalitarian/Xenophile build. Your patron protects you while you build relationships.

Galactic Empire Victory (Galactic Community)

Become the Galactic Emperor by passing the Galactic Empire reform resolution.

Aspect	Details
Best for	Authoritarian empires with strong diplomatic weight
Timeline	Years 60-200. Join early, build diplomatic weight, pass resolutions.
Key tech	Galactic Community formation, communications
Key resources	Diplomatic weight (fleet power + economy + population)
Mod impact	No major changes. AI mods may make voting more competitive.
Pitfall	You need fleet power for diplomatic weight — don't neglect your military just because you're going the diplomatic route.
Wave alignment	Wave 3 — After liberating yourself, dominate the GC to prevent your former overlord from retaliating.

Crisis Victory (Become the Crisis)

Unlock the Become the Crisis ascension perk and progress through its stages to build and fire the Aetherophasic Engine — destroying the galaxy.

Aspect	Details
Required DLC	Nemesis
Best for	Militarist/Xenophobe empires with strong alloy economy
Timeline	Years 100-250. Requires steady menace generation.
Key resources	Menace (earned through warfare, purging, planet cracking)
Key buildings	Menace buildings, Star-Eaters (planet-cracking ships)
Mod impact	Some mods add more toys for the Crisis path. Check compatibility.
Pitfall	Everyone will declare war on you at high menace levels. You need the strongest fleet in the galaxy.
Wave alignment	Post-curriculum content. Not recommended for learning playthroughs but a natural next challenge after Wave 3.

Score Victory

The default victory — highest score at victory year (2500 by default).

Aspect	Details
Best for	Any empire, especially tech/economy focused builds
Timeline	Continuous. Every decision affects your score.
Key factors	Population, technology, economy, fleet power, federations, subjects
Mod impact	No changes. Score is calculated the same way.
Pitfall	Score victory rewards passive play. You don't have to do anything dramatic to win on score.
Wave alignment	Wave 1 — your science victory path naturally wins on score if nothing else triggers.

4. Endgame Crisis Preparation

The end-game crisis is the final exam for every Stellaris campaign. There are four possible crises, each requiring a different counter-strategy.

Crisis Types

The Unbidden

Aspect	Details
Trigger	Researching jump drives repeatedly
Enemy type	Energy beings. Shields, shield-penetrating weapons.
Weakness	Armor-penetrating weapons (disruptors, arc emitters, cloud lightning). They have zero armor — bypassing shields kills them instantly.
Recommended fleet	Arc emitter battleships with carrier support (strike craft bypass shields).
Terrain	They spawn from a rift and expand outward. Rush the rift. Destroying it stops reinforcements.
Wave alignment	All waves — common crisis, no mod-specific adjustments needed.

The Contingency

Aspect	Details
Trigger	Mid-game year + certain conditions
Enemy type	AI robots. Armor-heavy, use energy weapons.
Weakness	Shield-penetrating weapons + hull damage. Kinetic weapons are effective against their shields; plasma/energy weapons melt their armor.
Recommended fleet	Balanced kinetic + energy. Artillery battleships + cruiser screen.
Terrain	Four machine worlds spawn in the galaxy. Each must be destroyed. A final hub world appears after all four fall.
Special	Sterilization hubs have system-wide effects. Don't engage without a strong fleet.
Mod impact	AI mods don't affect the Contingency directly. Gigastructures adds some related content.

The Prethoryn Scourge

Aspect	Details
Trigger	Mid-game year + random event
Enemy type	Organic swarm. Armor-heavy, use strike craft and missiles.
Weakness	Energy weapons + point defense. Their ships have high armor and no shields. Plasma, lasers, and disruptors are optimal. Point defense destroys their missiles and strike craft.
Recommended fleet	Plasma battleships + carrier cruisers with point defense.
Terrain	They arrive from the galactic rim, conquering inward. No central weak point — you must fight system by system.
Special	Prethoryn can be infected with a special project (research their biology) — this permanently debuffs them.
Mod impact	No major changes with standard modlist.

Cetana / The Crisis of Aeternum

Aspect	Details
Trigger	New crisis added in 4.4. Requires certain end-game conditions.
Enemy type	Unique crisis with its own mechanics.
Weakness	Varies by phase. Requires strong balanced fleet and repeatable techs.
Mod impact	Some mods may not have updated compatibility for Cetana. Check mod pages.

Crisis Preparation Timeline

Regardless of which crisis spawns, follow this economic preparation schedule:

Years Before Crisis	What to Do
100+ years	Build your economy. Specialize planets. Get your ascension path chosen. Build first megastructures.
50-100 years	Expand alloy production. Build a dedicated forge ecumenopolis or foundry habitat. Begin fleet buildup.
25-50 years	Fleet to 50% of your target crisis-fighting power. Research crisis-specific weapons. Build gateway network for rapid deployment.
10-25 years	Max out naval capacity. Build anchorages. Convert all spare industry to alloys. Set trade to militarized economy.
1-10 years	Fleet at target. Fortify border chokepoints. Starbases with communication jammers, target uplink computers. Rally fleets to a central gateway hub.
During crisis	Don't overcommit. Fight defensively. Use gateway network to respond to multiple fronts. Coordinate with AI empires (the GC crisis resolutions help).

Fleet Power Targets by Wave

Wave	Crisis Strength	Minimum Fleet Power	Comfortable Fleet Power
Wave 1	1x	100K	150K-200K
Wave 2	1x	150K	200K-300K
Wave 3	2x	250K	350K-500K

Crisis Preparation Checklist

- ☐ Ascension path chosen and fully unlocked
- ☐ Megastructure(s) operational — Science Nexus, Dyson Sphere, Matter Decompressor
- ☐ Gateway network connecting your core worlds
- ☐ Dedicated forge world(s) for alloy production
- ☐ Starbase bastions at border chokepoints (target uplink + communication jammer)
- ☐ Crisis-specific weapons researched
- ☐ Fleet at or above minimum power target
- ☐ Naval cap maxed out (anchorages, fortresses)
- ☐ Economy positive and stable (don't deficit-spend into a spiral during the crisis)
- ☐ Save game before unpausing — you may need to re-try the crisis fight

5. Megastructure Strategy

Megastructures define the late game. They provide resource streams that normal planets can't match.

Priority Order by Playstyle

Playstyle	1st Priority	2nd Priority	3rd Priority	Notes
Tech Rush	Science Nexus	Dyson Sphere	Mega Shipyard	Research first, then energy for upkeep, then ship production
Conquest	Mega Shipyard	Dyson Sphere	Matter Decompressor	Ships first — alloy income + ship build speed
Economic	Dyson Sphere	Matter Decompressor	Science Nexus	Energy -> minerals -> tech
Tall	Science Nexus	Dyson Sphere	Ring World	Research + energy + habitable space
Wide	Mega Shipyard	Matter Decompressor	Strategic Command Center	Ship production + minerals + fleet cap

Gigastructural Engineering Mod Integration

With Gigastructural Engineering active, the priority changes significantly:

Structure	What It Does	Build Priority
Alderson Disk	Massive energy production + habitable sections	High — but requires late-game tech
System Craft	Mobile planet-fortress	Medium — niche use for crisis fighting
Maginot World	Planetary defense fortress	Low — defensive-only
Quadrant Thermo Plant	Infinite energy from gas giants	Medium — good for energy-hungry empires
Planetcraft	Gigantic ship that one-shots fleets	Very high — but extremely expensive

Build Location Tips

- **Science Nexus:** Build in your home sector. Protects your research investment.
- **Dyson Sphere:** Build in a black hole system (if available) or an isolated system far from borders. They're vulnerable during construction.
- **Matter Decompressor:** Build on a black hole. No choice — it requires one.
- **Mega Shipyard:** Build near your front line (late game) or in your core sector (early game).
- **Ring World:** Build in a system with lots of empty space. Add gates for rapid population movement.

6. Tradition & Perk Choices

Traditions shape your empire from the early game to the end. The order you pick them matters more than the specific mix.

Recommended Order by Playstyle

Tech Rush (Wave 1)

1. **Discovery** — Faster research, anomaly discovery chance, survey speed. +1 research alternative.
2. **Expansion** — Cheaper starbases, faster colony development. More pops faster.
3. **Prosperity** — Cheaper buildings, better economy. +10% specialist output.
4. **Supremacy** — Fleet cap, fire rate. You need some defense.
5. **Adaptability** — Habitability, building slots. Helps with diverse planets.
6. **Diplomacy** — Envoys, federation cohesion. Useful for federation victory.
7. **Harmony** — Stability, happiness. Nice bonuses but lowest priority.

Key ascension perks: Technological Ascendancy, The Flesh is Weak/Synthetic Evolution, Master Builders, Galactic Wonders, Colossus Project.

Conquest (Wave 2)

1. **Supremacy** — Fleet cap, fire rate, ship build speed. War from day one.
2. **Domination** — Stability on conquered worlds, vassal bonuses, influence.
3. **Prosperity** — Alloy production, economy. War needs resources.
4. **Discovery** — You still need tech. Second tradition tree for research stations.
5. **Expansion** — More territory, more pops.
6. **Adaptability** — Habitability for conquered worlds of different climates.
7. **Diplomacy or Harmony** — Situational.

Key ascension perks: Defender of the Galaxy, Colossus Project, Master Builders, Galactic Wonders.

Survival (Wave 3 — Payback)

1. **Discovery** — Tech advantage essential. You start behind — catch up.
2. **Supremacy** — You must win your independence war.
3. **Prosperity** — Economy after the rebellion.
4. **Domination** — Vassal management, conquered territory.
5. **Expansion** — More planets, more fleet capacity.
6. **Adaptability** — 0 guaranteed worlds means weird planets.
7. **Harmony or Diplomacy** — Your choice.

Key ascension perks: Technological Ascendancy, Defender of the Galaxy, Galactic Force Projection (fleet cap when you need it most), Master Builders, Galactic Wonders.

Mod-Added Traditions

Some mods add new tradition trees. If you're playing with the full Wave 3 modlist, you may encounter:

- **Merged traditions mods** — Combine multiple trees. Read carefully before picking.
- **Expanded tradition mods** — Add new trees with unique bonuses.
- **Gigastructures traditions** — If Gigastructures adds a tradition, it's usually worth taking early for megastructure discounts.

7. Pacing Decisions

A Stellaris campaign is a 200-300 year arc. The empires that win know **when** to do things, not just **what** to do.

The Four Ages of Stellaris

Age	Years	What Matters	What to Avoid
Exploration	1-30	Surveying, anomalies, first contact, chokepoints, first colony	War. Don't fight — you have nothing to gain, everything to lose.
Expansion	30-100	Colonization, claims, vassals, starbase network, second/third colony ships	Overextension. Stop expanding when your economy can't support new colonies.
Consolidation	100-200	Specialization, megastructures, ascension completion, tech parity	Stagnation. If you stopped expanding, you must start building. Upgrade buildings.
Crisis	200+	Fleet combat, crisis mechanics, galaxy coordination	Debt. Don't deficit-spend alloys into a spiral. Keep one stable sector producing energy and food.

When to Expand vs When to Develop

Situation	Do This
You found a chokepoint system	Expand to it immediately. Claim it.
You found a 80%+ habitable planet	Colonize it. Even if it hurts short-term economy.
Your economy is negative in any resource	Pause expansion. Fix the deficit first.
You're at naval cap	Build more anchorages or fortresses. Don't waste fleet capacity.
You have 3+ undeveloped colonies	Stop colonizing. Build districts and buildings on what you have.
You're behind in tech (compared to neighbors)	Build research labs on all planets. Discovery tradition if you haven't picked it.
You're threatened by a neighbor	Build starbase bastion on the chokepoint. Supremacy tradition. Your fleet should match or exceed theirs.

When to Go to War

Situation	Go to War	Don't Go to War
Your fleet is stronger	Yes — strike before they build up	
Your fleet is weaker		No — build up first. Improve relations to buy time.
You can vassalize a small neighbor	Yes — one quick war for a long-term income boost	
You're already in a war		Finish your current war before starting another. Multi-front wars are a trap.
The crisis is coming in 50 years	Yes — conquer what you need before the crisis	
The crisis is here		No — all resources go to crisis defense. Don't start new wars during a crisis unless it's against the crisis.

Pacing by Wave

Wave 1: You're protected. Use this to learn the rhythm. Don't rush. Your patron absorbs the cost of your mistakes. Focus on the four ages in order — don't skip to the crisis phase before you've built your economy.

Wave 2: You must be aggressive early. Your first conquest needs to happen before year 2300. After that, alternate between consolidation (10-20 years) and war (5-10 years). Don't war constantly — your economy needs recovery time.

Wave 3: Your pacing is set by your overlord. Independence war around year 15-40. After freedom, you're in a catch-up phase (years 40-80). Then normal pacing. The early crisis (2375) means you have less time — start megastructures by year 2250 at the latest.

Quick Reference: Decision Flowchart

```
What's your situation?
├─ Just started (years 1-30)
│   ├── Survey everything
│   ├── Claim chokepoints
│   └─ Tech up (Discovery tradition)
├─ Established (years 30-100)
│   ├── Militarist? → Look for first conquest target
│   ├── Pacifist/Xenophile? → Build federation, explore
│   └─ Materialist? → Pile on research labs, rush key techs
├─ Strong mid-game (years 100-200)
│   ├── Pick your ascension path if you haven't
│   ├── Start first megastructure
│   ├── Build gateway network
│   └─ Crisis prep: alloys, fleet, starbases
└─ Crisis imminent (years 200+)
    ├── Identify crisis type
    ├── Counter-build fleet
    └─ Mobilize galaxy via GC
```

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Performance

Work in progress. This page will document empirical performance benchmarks comparing vanilla vs. modded Stellaris.

Methodology

- **Hardware:** [Your specs here]
- **Settings:** 1920x1080, Ultra quality, Fastest galaxy
- **Measurement tool:** FPS counter (Steam overlay), task manager for load times
- **Save:** 100-year autosave from a standard game

Benchmark Results

Scenario	FPS (min/avg)	Load Time	Notes
Vanilla	—	—	Baseline
All mods	—	—	Full modlist
Graphics mods only	—	—	Visual comparison
Gameplay mods only	—	—	Gameplay comparison

Known Performance Impacts

- **Graphics mods:** High-res textures and lighting overhauls can cost 5-15 FPS depending on GPU
- **Pop-heavy mods:** Mods that add many new pops or jobs will slow late-game
- **UI mods:** Minimal to no impact
- **Galaxy size mods:** Larger galaxies increase load times and late-game lag

Tips for Better Performance

- Reduce galaxy size from Medium to Small
- Lower hyperlane density
- Set planet habitability to lower values (fewer colonization targets)
- Disable xeno-compatibility if it's in your modlist (notorious for pop bloat)
- Use the "Pop Scaling" option in Stellaris game settings

Missing Mods (Research)

Research & planning file. Mods that don't exist yet but would complete the modlist vision. These are candidates for custom development.

Mod Idea	Category	Why It's Needed	Priority	Notes
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Appendix

Glossary

Term	Definition
Origin	Your empire's starting condition and backstory
Hyperlane	The travel routes between star systems
Starbase	Your control point in each system; allows ship building and defense
Pop	Population unit on a planet; each pop works a job
District	Planet area type (Energy, Mining, Agriculture, Industrial, City)
Building	Planet structure that provides advanced jobs
Megastructure	Large-scale construction project (Dyson Sphere, Matter Decompressor, etc.)
Crisis	End-game threat that emerges around 2400
Ascension Path	End-game tradition tree (Psionic, Genetic, or Synthetic)
Federation	Alliance of empires with shared rules and benefits
Galactic Community	Galactic-wide voting body that passes laws
Edict	Empire-wide modifier you activate with influence or unity

Troubleshooting

Game crashes on launch

- Disable all mods. Re-enable one by one to find the culprit.
- Check MO2's log for error messages.
- Verify Stellaris game files in Steam.

Mods not showing in game

- Ensure MO2 profile is correct.
- Make sure mods are enabled in the left panel.
- Check that the mod's game version matches your Stellaris version.

Performance issues

- See [Performance](#) for optimization tips.
- Disable graphics-intensive mods if late-game lag persists.

Save won't load

- You removed a mod that the save depends on. Re-enable it.
- A mod updated mid-campaign. Roll back to the previous version (MO2 version locking prevents this).

Credits

Mod Authors

All mods in this guide belong to their respective creators. Links to each mod's source are provided in the modlist sections.

Guide Tools

- [MkDocs](#) — site generation
- [Material for MkDocs](#) — theme
- [mkdocs-exporter](#) — PDF export
- [Mod Organizer 2](#) — mod management